

Tsung-Wei (TW) Huang

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Appointment

Associate Professor, Department of Electrical and Computer Engineering 2025 – present
University of Wisconsin at Madison, Madison, Wisconsin, USA

Assistant Professor, Department of Electrical and Computer Engineering 2023 – 2025
University of Wisconsin at Madison, Madison, Wisconsin, USA

Assistant Professor, Department of Electrical and Computer Engineering 2019 – 2023
University of Utah, Salt Lake City, Utah, USA

Research Assistant Professor, Department of Electrical and Computer Engineering 2018 – 2019
University of Illinois at Urbana-Champaign, IL, USA

Education

Postdoc, Department of Electrical and Computer Engineering 2018 – 2018
University of Illinois at Urbana-Champaign, IL, USA

PhD, Department of Electrical and Computer Engineering 2013 – 2017
University of Illinois at Urbana-Champaign, IL, USA

MS, Department of Computer Science and Information Engineering 2010 – 2011
National Cheng Kung University, Tainan, Taiwan

BS, Department of Computer Science and Information Engineering 2006 – 2010
National Cheng Kung University, Tainan, Taiwan

Research Interest

Electronic Design Automation, High-performance Computing, Quantum Computing

Software Project

Our software projects have been used by thousands of people from both industry and academia:

1. Taskflow: A General-purpose Parallel and Heterogeneous Programming System
https://taskflow.github.io/ (8–11K weekly downloads)

- C++ Conference Best Poster Award, 2025
- PPoPP FastCode Programming Challenge Winner (2nd Place), 2025
- MIT/Amazon/HPEC Graph Challenge Innovation Award (2nd Place), 2023
- MIT/Amazon/HPEC Graph Challenge Champion Award (1st Place), 2020
- ACM Multimedia Best Open-source Software Award, 2019
- C++ Conference Best Poster Award, 2018

2. OpenTimer: A High-performance Timing Analysis Tool for VLSI Systems
https://github.com/OpenTimer/OpenTimer

- IEEE/ACM ICCAD 10-year Retrospective Most Influential Paper Award, 2024
- ACM SIGDA Outstanding PhD Dissertation Award, 2019
- Best EDA Software Tool, WOSET@ICCAD, 2018
- Top-3 Winners of ACM TAU Contests, 2014–2016
- Golden Timers of ACM TAU Contests, 2017–2021
- Golden Timer of IEEE/ACM ICCAD CAD Contest, 2015

3. RTLflow: A GPU Acceleration Flow for RTL Simulation with Batch Stimulus
https://github.com/dian-lun-lin/rtlflow

4. SNIG: A Task-parallel Inference Engine for Large Sparse Neural Network
https://github.com/dian-lun-lin/SNIG

- MIT/Amazon/HPEC Graph Challenge Champion Award, 2020

5. DtCraft: A Data-parallel Distributed Streaming System
https://github.com/twhuang-uiuc/DtCraft

- ACM Multimedia Best Open-source Software Award, 2018

AWARDS

- Ernest S. Kuh Early Career Award, IEEE Council of Electronic Design Automation (CEDA), 2026
- Open Award Hall of Fame, Open Source Program Office (OSPO) at UW-Madison, 2026
- Outstanding Paper Award (5 out of 145 papers), IEEE HPEC, 2026
- Vilas Early-career Investigator Award, Trustees of the William F. Vilas Estate. Funding, 2025
- DAC Under-40 Innovator Award, 2025
- CoE Award for Excellence in Graduate Student Mentoring, UW-Madison, 2025
- Second Place, ACM SIGPLAN PPoPP Fast Code Programming Challenge (FCPC), 2025
- Honorable Mention (6th Place out of 90 teams), IEEE/ACM ICCAD CAD Contest, 2024
- ICCAD 10-year Retrospective Most Influential Paper Award, 2024
- 2nd Place, MIT/Amazon/HPEC Large Sparse Neural Network Challenge, 2023
- 2nd Place, MIT/Amazon/HPEC Streaming Graph Challenge, 2023
- ACM SIGDA Outstanding New Faculty Award, 2023
- ACM SIGDA Meritorious Service Award, 2022
- Humboldt Research Fellowship Award, Alexander von Humboldt Foundation, 2022
- Faculty Early Career Development Program (CAREER) Award, NSF, 2022
- Best Paper Award for "GPU-Accelerated Path-based Timing Analysis," ACM TAU Workshop, 2021
- 1st Place, MIT/Amazon/HPEC Large Sparse Neural Network Challenge, 2020
- 2nd Place (Taskflow), Open-source Software Competition, ACM Multimedia Conference, 2019
- ACM SIGDA Outstanding PhD Dissertation Award (thesis title: "Distributed Timing Analysis"), 2019
- Best Tool Award (OpenTimer), Workshop on Open-source EDA Technology, 2018
- Best Open-source Software Award (DtCraft), ACM Multimedia Conference, 2018
- Best Poster Award (Taskflow), CPP Conference, 2018
- 2nd and 1st Place, ACM/SIGDA CADathlon International Programming Contest, 2014 and 2017
- 1st, 2nd, and 1st Place, ACM TAU Timing Analysis Contest, 2014–2016
- Yi-Min Wang and Pi-Yu Chung Endowed Research Award, ECE Dept. UIUC, 2016
- Rambus Computer Engineering Fellowship, ECE Dept. UIUC, 2015—2016
- Study Abroad Scholarship, Ministry of Education, Taiwan, 2013—2014
- 2nd Place, ACM Student Research Competition Grand Final, ACM Annual Award Banquet, 2011
- Best Master's Thesis Award, Taiwan Institute of Electrical and Electronic Engineering, 2011
- Best Master's Thesis Award, IEEE Taiwan Tainan Section, 2011
- Best Master's Thesis Award, Taiwan Institute of Information and Computing Machinery, 2011
- 1st Place, Master's Thesis Contest, Chinese Institute of Electrical Engineering, Taiwan, 2011
- Outstanding Graduate Recruiting Fellowship, National Cheng Kung University, 2010
- Outstanding Student Scholarship, Garmin Corporation, Taiwan, 2010
- 1st Place, ACM/SIGDA Student Research Competition, Design Automation Conference, 2010
- 3rd Place, National Collegiate Cell-Based IC Design Contest, Ministry of Education, Taiwan, 2010
- Distinguished Engineering Student Fellowship, Chinese Institute of Engineers, Taiwan, 2009

- 1st Place, National Collegiate Nano Device CAD Contest, Nano Device Laboratories, Taiwan, 2009
- 3rd Place, National Collegiate Programming Contest, Ministry of Education, Taiwan, 2009
- 2nd Place, National Collegiate IC/CAD Programming Contest, Ministry of Education, Taiwan, 2009
- 2nd Place, Presidential Award in CS Department, National Cheng Kung University, Taiwan, 2009

Research Grants

Fall Research Competition Award
PI, $56K, Office of the Vice Chancellor for Research, UW-Madison — July 2026 – June 2027

Vilas Early-career Investigator Award
PI, $60K, Trustees of the William F. Vilas Estate. Funding — July 2025 – July 2028

Co-Design of Chiral Quantum Photonic Devices and Circuits
Co-PI, $400K, NSF, DMR-2235276 — Aug 2023 – July 2025

Toward a Task-parallel Programming Ecosystem for Modern Scientific Computing
PI, $298K, NSF, TI-2229304/-2349144 — Sep 2022 – Aug 2024

GPU Acceleration for Satisfiability Solver
PI, $5K (hardware donation), Intel — Oct 2022

Developer Training Programs for Taskflow
PI, $5K, NumFOCUS Small Development Grant — Sep 2022 – May 2023

Transpiling Parallel Task Graph Programming Models for Scientific Software
PI, $488K, NSF, OAC-2209957/-2349143 — July 2022 – July 2025

Taskflow with Constrained Parallelism
PI, $16K, NSF, CCF-2126672 (REU supplement) — Aug 2022 – Aug 2023

Accelerating Static Timing Analysis with Intelligent Heterogeneous Parallelism
PI, $500K, NSF, CCF-2144523/-2349582 (CAREER) — Jan 2022 – Jan 2027

GPU Acceleration for Static Timing Analysis
PI, $10K (hardware donation), Nvidia Applied Research Acceleration Program — Nov 2021

A General-purpose Heterogeneous Task Graph Computing System for VLSI CAD
PI, $403K, NSF, CCF-2126672/-2349141 — Oct 2021 – Oct 2024

Standard GPU Algorithms with Task Graph Parallelism
PI, $5K, NumFOCUS Small Development Grant — May 2021 – Feb 2022

Taskflow-San: Sanitizing Erroneous Control Flows in Taskflow
PI, $5K, NumFOCUS Small Development Grant — May 2021 – Feb 2022

OpenTimer and DtCraft
PI, $427K, DARPA, FA 8650-18-2-7843 — June 2018 – July 2019

Conference Publication

1. **[ASPLOS'27]** Yi-Hua Chung, Hao-Hsiang Hsiao, Cheng-Hsiang Chiu, Boyang Zhang, Che Chang, Joshua San Miguel, and Tsung-Wei Huang, "XEG: A GPU-Parallel Algorithm for Efficient and Effective E-Graph Extraction," *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Heraklion, Crete, Greece, 2027
2. **[CGO'27]** Chih-Chun Chang, Aditya Das Sarma, Cheng-Hsiang Chiu, and Tsung-Wei Huang, "Can ML Frameworks Go Beyond ML? A Case Study in GPU-accelerated VLSI Timing Analysis," *IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, Salt Lake City, UT, 2027
3. **[PACT'26]** Shui Jiang, Hengrui Chen, Tsung-Yi Ho, and Tsung-Wei Huang, "Q-TranSim: Batch Quantum Circuit Simulation using Tensor Transpilation," *ACM International Conference on Parallel Architectures and Compilation Techniques (PACT)*, Chicago, Illinois, 2026

4. **[MICRO'26]** Xiangyu Ren, Yuexun Huang, Zhaohui Yang, Yuchen Zhu, Tsung-Wei Huang, Tsung-Yi Ho, Zhiding Liang, and Antonio Barbalace, "A Spatial-Topology Preserving Compiler for Quantum Many-Body Systems Simulation," *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, Athens, Greece, 2026
5. **[Euro-Par'26]** Boyang Zhang, Chih-Chun Chang, Yi-Hua Chung, Che Chang, Cheng-Hsiang Chiu, Aditya Das Sarma, and Tsung-Wei Huang, "G-STAR: GPU-Accelerated Statistical Static Timing Analysis using Level-by-level Replication," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, Pisa, Italy, 2026
6. **[Euro-Par'26]** Zhengxiong Li, Tsung-Wei Huang, and Umit Ogras, "SET: Stream-Event-Triggered Scheduling for Efficient CUDA Graph Pipelines," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, Pisa, Italy, 2026
7. **[ICS'26]** Che Chang, Yi-Hua Chung, Cheng-Hsiang Chiu, Wan-Luan Lee, Boyang Zhang, Ulf Schlichtmann, Ing-Chao Lin, Xiangyao Yu, and Tsung-Wei Huang, "G-PathGen: An Efficient GPU-Parallel k-Critical Path Generation Algorithm," *ACM International Conference on Supercomputing (ICS)*, Belfast, Northern Ireland, UK, 2026
8. **[DAC'26]** Aditya Das Sarma, Wan-Luan Lee, Shui Jiang, Boyang Zhang, and Tsung-Wei Huang, "A Differentiable Approach to Task Graph Partitioning: A Case Study in RTL Simulation," *ACM/IEEE Design Automation Conference (DAC)*, Long Beach, CA, 2026
9. **[DAC'26]** Wan-Luan Lee, Aditya Das Sarma, Che Chang, Chih-Chun Chang, and Tsung-Wei Huang, "iHyperG: Incremental Hypergraph Partitioning on GPU," *ACM/IEEE Design Automation Conference (DAC)*, Long Beach, CA, 2026
10. **[DAC'26]** Pingchuan Ma, Ziang Yin, Qi Jing, Zhengqi Gao, Nicholas Gangi, Boyang Zhang, Tsung-Wei Huang, Rena Huang, Duane Boning, Yu Yao, and Jiaqi Gu, "SP2RINT: Spatially-Decoupled Physics-Constrained Progressive Inverse Optimization for Diffractive Optical Neural Network Training," *ACM/IEEE Design Automation Conference (DAC)*, Long Beach, CA, 2026
11. **[ASP-DAC'26]** Aditya Das Sarma, Shui Jiang, Wan-Luan Lee, Tsung-Yi Ho, and Tsung-Wei Huang, "TIMBER: A Fast Algorithm for Timing and Power Optimization using Multi-bit Flip-flops," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Hong Kong, 2026
12. **[HPEC'25]** Chedi Morchdi, Cheng-Hsiang Chiu, Wan-Luan Lee, Tsung-Wei Huang, and Yi Zhou, "Enhancing Graph Partitioning with Reinforcement Learning-based Initialization," *IEEE High-performance and Extreme Computing Conference (HPEC)*, virtual, 2025
13. **[HPEC'25]** Jie Tong, Zhengxiong Li, Umit Yusuf Ogras, and Tsung-Wei Huang, "A Scalable Code Generation Flow for Heterogeneous Parallel RTL Simulation using MLIR," *IEEE High-performance and Extreme Computing Conference (HPEC)*, virtual, 2025
14. **[MLCAD'25]** Cheng-Hsiang Chiu, Chedi Morchdi, Chih-Chun Chang, Cunxi Yu, Yi Zhou, and Tsung-Wei Huang, "Optimizing CUDA Graph Scheduling with Reinforcement Learning: A Case Study in SSTA Propagation," *ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)*, San Cruz, CA, 2025
15. **[Euro-Par'25]** Yi-Hua Chung, Shui Jiang, Wan Luan Lee, Yanqing Zhang, Haoxing Ren, Tsung-Yi Ho, and Tsung-Wei Huang, "SimPart: A Simple Yet Effective Replication-aided Partitioning Algorithm for Logic Simulation on GPU," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, Dresden, Germany, 2025
16. **[Euro-Par'25]** Jie Tong, Wan-Luan Lee, Umit Yusuf Ogras, and Tsung-Wei Huang, "Scalable Code Generation for RTL Simulation of Deep Learning Accelerators with MLIR," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, Dresden, Germany, 2025
17. **[DAC'25]** Wan-Luan Lee, Shui Jiang, Dian-Lun Lin, Che Chang, Boyang Zhang, Yi-Hua Chung, Ulf Schlichtmann, Tsung-Yi Ho, and Tsung-Wei Huang, "iG-kway: Incremental k-way Graph Partitioning on GPU," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2025
18. **[DAC'25]** Chih-Chun Chang and Tsung-Wei Huang, "Statistical Timing Graph Scheduling Algorithm for GPU Computation," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2025

19. **[ASPLOS'25]** Shui Jiang, Yi-Hua Chung, Chih-Chun Chang, Tsung-Yi Ho, and Tsung-Wei Huang, "BQSim: GPU-accelerated Batch Quantum Circuit Simulation using Decision Diagram," *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Rotterdam, Netherlands, 2025
20. **[FCPC'25]** Pao-I Chen and Tsung-Wei Huang, "An Efficient Implemenetation of Parallel Breadth-first Search," *ACM SIGPLAN PPoPP Workshop on Fast Code Programming Challenge (FCPC)*, Las Vegas, Nevada, 2025
21. **[ExHET'25]** Serhan Gener, Sahil Hassan, Liangliang Chang, Chaitali Chakrabarti, Tsung-Wei Huang, Umit Ograss, and Ali Akoglu, "A Unified Portable and Programmable Framework for Task-Based Execution and Dynamic Resource Management on Heterogeneous Systems," *ACM International Workshop on Extreme Heterogeneity Solutions (ExHET)*, Las Vegas, Nevada, 2025
22. **[WAMTA'25]** Cheng-Hsiang Chiu, Wan-Luan Lee, Boyang Zhang, Yi-Hua Chung, Che Chang, and Tsung-Wei Huang, "A Task-parallel Pipeline Programming Model with Token Dependency," *Workshop on Asynchronous Many-Task Systems and Applications (WAMTA)*, St. Louis, MO, 2025
23. **[ASP-DAC'25]** Wan-Luan Lee, Dian-Lun Lin, Cheng-Hsiang Chiu, Ulf Schlichtmann, and Tsung-Wei Huang, "HyperG: Multilevel GPU-Accelerated k-way Hypergraph Partitioner," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan, 2025
24. **[ASP-DAC'25]** Boyang Zhang, Che Chang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yang Sui, Chih-Chun Chang, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "iTAP: An Incremental Task Graph Partitioner for Task-parallel Static Timing Analysis," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan, 2025
25. **[ASP-DAC'25]** Che Chang, Boyang Zhang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "PathGen: An Efficient Parallel Critical Path Generation Algorithm," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan, 2025
26. **[ASP-DAC'25]** Cheng-Hsiang Chiu, Chedi Morchdi, Yi Zhou, Boyang Zhang, Che Chang, and Tsung-Wei Huang, "Reinforcement Learning-generated Topological Order for Dynamic Task Graph Scheduling," *IEEE High-performance and Extreme Computing Conference (HPEC)*, MA, 2024
27. **[ICCAD'25]** Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, Yufan Du, Yibo Lin, Runsheng Wang, and Ru Huang, "HeteroExcept: Heterogeneous Engine for General Timing Path Exception Analysis," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, New York, 2024
28. **[ICPP'24]** Chih-Chun Chang, Boyang Zhang, and Tsung-Wei Huang, "GSAP: A GPU-Accelerated Stochastic Graph Partitioner" *ACM International Conference on Parallel Processing (ICPP)*, Gotland, Sweden, 2024
29. **[ICPP'24]** Shui Jiang, Rongliang Fu, Lukas Burgholzer, Robert Wille, Tsung-Yi Ho, and Tsung-Wei Huang, "FlatDD: A High-Performance Quantum Circuit Simulator using Decision Diagram and Flat Array," *ACM International Conference on Parallel Processing (ICPP)*, Gotland, Sweden, 2024
30. **[Euro-Par'24]** Dian-Lun Lin, Joshua San Miguel, Umit Ogras, Tsung-Wei Huang, "TaroRTL: Accelerating RTL Simulation using Coroutine-based Heterogeneous Task Graph Scheduling," *International European Conference on Parallel and Distributed Computing (Euro-Par)*, Madrid, Spain, 2024
31. **[DAC'24]** Wan Luan Lee, Dian-Lun Lin, Tsung-Wei Huang, Shui Jiang, Tsung-Yi Ho, Yibo Lin, and Bei Yu, "G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2024
32. **[DAC'24]** Che Chang, Tsung-Wei Huang, Dian-Lun Lin, Guannan Guo, and Shiju Lin, "Ink: Efficient Incremental k-Critical Path Generation," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2024
33. **[DAC'24]** Boyang Zhang, Dian-Lun Lin, Che Chang, Cheng-Hsiang Chiu, Bojue Wang, Wan Luan Lee, Chih-Chun Chang, Donghao Fang, and Tsung-Wei Huang, "G-PASTA: GPU Accelerated Partitioning Algorithm for Static Timing Analysis," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2024

34. **[DAC'24]** Shiju Lin, Guannan Guo, Tsung-Wei Huang, Weihua Sheng, Evangeline Young, and Martin Wong, "GCS-Timer: GPU-Accelerated Current Source Model Based Static Timing Analysis," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2024
35. **[ISVLSI'24]** Jie Tong, Liangliang Chang, Umit Yusuf Ogras, and Tsung-Wei Huang, "BatchSim: Parallel RTL Simulation using Inter-cycle Batching and Task Graph Parallelism," *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Knoxville, Tennessee, 2024
36. **[ISVLSI'24]** Che Chang, Cheng-Hsiang Chiu, Boyang Zhang, and Tsung-Wei Huang, "Incremental Critical Path Generation for Dynamic Graphs," *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Knoxville, Tennessee, 2024
37. **[ISVLSI'24]** Cheng-Hsiang Chiu and Tsung-Wei Huang, "An Experimental Study of Dynamic Task Graph Parallelism for Large-Scale Circuit Analysis Workloads," *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Knoxville, Tennessee, 2024
38. **[AAMAS'24]** Shao-Hung Chan, Zhe Chen, Dian-Lun Lin, Yue Zhang, Daniel Harabor, Tsung-Wei Huang, Sven Koenig, and Thomy Phan, "Anytime Multi-Agent Path Finding using Operator Parallelism in Large Neighborhood Search," *International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, Auckland, New Zealand, 2024
39. **[ISPD'24]** Tsung-Wei Huang, Boyang Zhang, Dian-Lun Lin, and Cheng-Hsiang Chiu, "Parallel and Heterogeneous Timing Analysis: Partition, Algorithm, and System," *ACM International Symposium on Physical Design (ISPD)*, Taipei, Taiwan, 2024
40. **[HPC-Asia]** Cheng-Hsiang Chiu, Zhicheng Xiong, Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "An Efficient Task-parallel Pipeline Programming Framework," *ACM International Conference on High-performance Computing in Asia-Pacific Region (HPC-Asia)*, Nagoya, Japan, 2024
41. **[DATE'24]** Zizheng Guo, Tsung-Wei Huang, Jin Zhou, Cheng Zhuo, Yibo Lin, Runsheng Wang, and Ru Huang, "Heterogeneous Static Timing Analysis with Advanced Delay Calculator," *IEEE/ACM Design, Automation and Test in Europe Conference (DATE)*, Valencia, Spain, 2024
42. **[ASP-DAC'24]** Chedi Morchdi, Cheng-Hsiang Chiu, Yi Zhou, and Tsung-Wei Huang, "A Resource-efficient Task Scheduling System using Reinforcement Learning," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Korea, 2024
43. **[ICCAD'23]** Cheng-Hsiang Chiu, Dian-Lun Lin, and Tsung-Wei Huang, "Programming Dynamic Task Parallelism for Heterogeneous EDA Algorithms," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Diego, 2023
44. **[ICCAD'23]** Takashi Sato, Chun-Yao Wang, Yu-Guang Chen, and Tsung-Wei Huang, "Overview of 2023 CAD Contest at ICCAD," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Diego, 2023
45. **[HPEC'23]** Shui Jiang, Tsung-Wei Huang, and Tsung-Yi Ho, "GLARE: Accelerating Sparse DNN Inference Kernels with Global Memory Access Reduction," *IEEE High Performance Extreme Computing (HPEC)*, Virtual, 2023 (**Graph Challenge Innovation Award**)
46. **[HPEC'23]** Chih-Chun Chang and Tsung-Wei Huang, "GLARE: Accelerating Sparse DNN Inference Kernels with Global Memory Access Reduction," *IEEE High Performance Extreme Computing (HPEC)*, Virtual, 2023 (**Graph Challenge Innovation Award**)
47. **[ICPP'23]** Shui Jiang, Tsung-Wei Huang, Bei Yu, and Tsung-Yi Ho, "SNICIT: Accelerating Sparse Neural Network Inference via Compression at Inference Time on GPU," *ACM International Conference on Parallel Processing (ICPP)*, Salt Lake City, Utah, 2023
48. **[DAC'23]** Dian-Lun Lin, Yanqing Zhang, Haoxing Ren, Shih-Hsin Wang, Brucek Khailany, and Tsung-Wei Huang, "GenFuzz: GPU-accelerated Hardware Fuzzing using Genetic Algorithm with Multiple Inputs," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2023
49. **[IPDPS'23]** Tsung-Wei Huang, "qTask: Task-parallel Quantum Circuit Simulation with Incrementality," *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, St. Petersburg, Florida, 2023
50. **[IPDPSW'23]** Elmir Dzaka, Dian-Lun Lin, and Tsung-Wei Huang, "Parallel And-Inverter Graph Simulation Using a Task-graph Computing System," *IEEE International Parallel and Distributed Processing Symposium Workshop (IPDPSW)*, St. Petersburg, Florida, 2023

51. **[DATE'23]** Guannan Guo, Martin D. F. Wong, and Tsung-Wei Huang, "Fast STA Graph Partitioning Framework for Multi-GPU Acceleration," *IEEE/ACM Design, Automation and Test in Europe Conference (DATE)*, Antwerp, Belgium, 2023
52. **[HPEC'22]**Tsung-Wei Huang and Leslie Hwang, "Task-parallel Programming with Constrained Parallelism," *IEEE High-performance Extreme Computing (HPEC)*, Waltham, MA, 2022
53. **[HPEC'22]** Tsung-Wei Huang, "Enhancing the Performance Portability of Heterogeneous Circuit Analysis Programs," *IEEE High-performance Extreme Computing (HPEC)*, Waltham, MA, 2022
54. **[ICPP'22]** Dian-Lun Lin, Haoxing Ren, Yanqing Zhang, Brucek Khailany, and Tsung-Wei Huang, "From RTL to CUDA: A GPU Acceleration Flow for RTL Simulation with Batch Stimulus," *ACM International Conference on Parallel Processing (ICPP)*, Bordeaux, France, 2022
55. **[HPDC'22]** Cheng-Hsiang Chiu and Tsung-Wei Huang, "Composing Pipeline Parallelism using Control Taskflow Graph," *ACM International Symposium on High-Performance Parallel and Distributed Computing (HPDC)*, Minneapolis, Minnesota, 2022
56. **[ICCAD'22]** Yu-Guan Chen, Chun-Yao Wang, Tsung-Wei Huang, and Takashi Sato, "Overview of 2022 CAD Contest at ICCAD," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Diego, CA, 2022
57. **[DAC'22]** Cheng-Hsiang Chiu and Tsung-Wei Huang, "Efficient Timing Propagation with Simultaneous Structural and Pipeline Parallelisms," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2022
58. **[HIPS'22]** Tsung-Wei Huang and Yibo Lin, "Concurrent CPU-GPU Task Programming using Modern C++," *International Workshop on High-Level Parallel Programming Models and Supportive Environments (HIPS)*, France, 2022
59. **[ASP-DAC'22]** Kexing Zhou, Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "Efficient Critical Paths Search Algorithm using Mergeable Heap," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Taiwan, 2022
60. **[DAC'21]** Guannan Guo, Tsung-Wei Huang, and Martin Wong, "GPU-accelerated Path-based Timing Analysis," *ACM/IEEE Design Automation Conference (DAC)*, CA, 2021
61. **[DAC'21]** Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "A Provably Good and Practically Efficient Common Path Pessimism Removal Algorithm for Large Designs," *ACM/IEEE Design Automation Conference (DAC)*, CA, 2021
62. **[ESPM2'21]** McKay Mower, Luke Majors, and Tsung-Wei Huang, "Taskflow-San: Sanitizing Erroneous Control Flow in Taskflow Programs," *IEEE Workshop on Extreme Scale Programming Models and Middleware (ESPM2)*, St. Louis, Missouri, 2021
63. **[ProTools'21]** Tsung-Wei Huang, "TFProf: Profiling Large Taskflow Programs with Modern D3 and C++," *IEEE International Workshop on Programming and Performance Visualization Tools (ProTools)*, St. Louis, Missouri, 2021
64. **[Euro-Par'21]** Dian-Lun Lin and Tsung-Wei Huang, "Efficient GPU Computation using Task Graph Parallelism," *European Conference on Parallel and Distributed Computing (Euro-Par)*, Portugal, 2021
65. **[HPEC'21]** Yasin Zamani and Tsung-Wei Huang, "A High-Performance Heterogeneous Critical Path Analysis Framework," *IEEE High-performance Extreme Computing (HPEC)*, Waltham, MA, 2021
66. **[AMET'21]** Cheng-Hsiang Chiu, Dian-Lun Lin and Tsung-Wei Huang, "An Experimental Study of SYCL Task Graph Parallelism for Large-Scale Machine Learning Workloads," *International Workshop of Asynchronous Many-Task Systems for Exascale (AMTE)*, 2021
67. **[ICCAD'21]** Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "HeteroCPPR: Accelerating Common Path Pessimism Removal with Heterogeneous CPU-GPU Parallelism," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, Germany, 2021
68. **[ICCAD'21]** Guannan Guo, Tsung-Wei Huang, Yibo Lin, and Martin D. F. Wong, "GPU-accelerated Critical Path Generation with Path Constraints," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, Germany, 2021

69. **[ICCAD'21]** Tsung-Wei Huang, Yu-Guan Chen, Chun-Yao Wang, and Takashi Sato, "Overview of 2021 CAD Contest at ICCAD," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, Germany, 2021
70. **[ASP-DAC'21]** Kuan-Ming Lai, Tsung-Wei Huang, Pei-Yu Lee, and Tsung-Yi Ho, "ATM: A High Accuracy Extracted Timing Model for Hierarchical Timing Analysis," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan, 2021
71. **[ICPADS'21]** Chun-Xun Lin, Tsung-Wei Huang, and Martin D. F. Wong, "An Efficient Work-Stealing Scheduler for Task Dependency Graph," *IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, Hong Kong, 2020
72. **[HPEC'20]** Dian-Lun Lin and Tsung-Wei Huang, "A Novel Inference Algorithm for Large Sparse Neural Network using Task Graph Parallelism," *IEEE High-performance Extreme Computing (HPEC)*, Waltham, MA, 2020 (**Graph Challenge Champion Award**)
73. **[ICCAD'20]** Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "GPU-Accelerated Static Timing Analysis," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Diego, 2020
74. **[ICCAD'20]** Tsung-Wei Huang, "A General-purpose Parallel and Heterogeneous Task Programming System for VLSI CAD," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Diego, 2020
75. **[ICCAD'20]** Ing-Chao Lin, Ulf Schlichtmann, Tsung-Wei Huang, and Pao-Hun Lin, "Overview of 2020 CAD Contest at ICCAD," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Diego, 2020
76. **[DAC'20]** Guannan Guo, Tsung-Wei Huang, Chun-Xun Lin, and Martin D. F. Wong, "An Efficient Critical Path Generation Algorithm Considering Extensive Path Constraints," *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, 2020
77. **[MM'19]** Chun-Xun Lin, Tsung-Wei Huang, Guannan Guo, and Martin D. F. Wong, "A Modern C++ Parallel Task Programming Library," *ACM Multimedia Conference (MM)*, Nice, France, 2019 (**Second Prize of Open-Source Software Competition**)
78. **[HPEC'19]** Chun-Xun Lin, Tsung-Wei Huang, Guannan Guo, and Martin D. F. Wong, "An Efficient and Composable Parallel Programming Library," *IEEE High-performance Extreme Computing (HPEC)*, Waltham, MA, 2019
79. **[IPDPS'19]** Tsung-Wei Huang, Chun-Xun Lin, Guannan Guo, and Martin D. F. Wong, "Cpp-Taskflow: Fast Task-based Parallel Programming using Modern C++," *IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, Rio De Janeiro, Brazil, 2019
80. **[DAC'19]** Kuan-Ming Lai, Tsung-Wei Huang, and Tsung-Yi Ho, "A General Cache Framework for Efficient Generation of Timing Critical Paths," *ACM/IEEE Design Automation Conference (DAC)*, Las Vegas, NV, 2019
81. **[DAC'19]** Tsung-Wei Huang, Chun-Xun Lin, Guannan Guo, and Martin D. F. Wong, "Essential Building Blocks for Creating an Open-source EDA Project," *ACM/IEEE Design Automation Conference (DAC)*, Las Vegas, NV, 2019
82. **[DAC'19]** Tsung-Wei Huang, Chun-Xun Lin, and Martin D. F. Wong, "Distributed Timing Analysis at Scale," *ACM/IEEE Design Automation Conference (DAC)*, Las Vegas, NV, 2019
83. **[MM'18]** Tsung-Wei Huang, Chun-Xun Lin, Guannan Guo, and Martin D. F. Wong, "A General-purpose Distributed Programming Systems using Data-parallel Streams," *ACM Multimedia Conference (MM)*, Seoul, Korea, 2018 (**Best Open-Source Software Award**)
84. **[DEBS'18]** Chun-Xun Lin, Tsung-Wei Huang, Guannan Guo, and Martin D. F. Wong, "MtDetector: A High-performance Marine Traffic Detector at Stream Scale," *ACM Distributed Event-based System Conference (DEBS)*, Hamilton, New Zealand, 2018
85. **[GLSVLSI'18]** Chun-Xun Lin, Tsung-Wei Huang, T. Yu, and Martin D. F. Wong, "A Distributed Power Grid Analysis Framework from Sequential Stream Graph," *ACM Great Lakes Symposium (GLSVLSI)*, Chicago, IL, 2018
86. **[ISQED'18]** Chun-Xun Lin, Tsung-Wei Huang, and Martin D. F. Wong, "Routing at Compile Time," *IEEE International Symposium on Quality Electronic Design (ISQED)*, Santa Clara, CA, 2018

87. **[ICCAD'17]** Tsung-Wei Huang, Chun-Xun Lin, and Martin D. F. Wong, "DtCraft: A Distributed Execution Engine for Compute-intensive Applications," *ACM/IEEE International Conference on Computer-aided Design (ICCAD)*, Irvine, CA, 2017
88. **[DAC'17]** Tin-Yin Lai, Tsung-Wei Huang, and Martin D. F. Wong, "An Effective and Accurate Macro-modeling Algorithm for Large Hierarchical Designs," *ACM/IEEE Design Automation Conference (DAC)*, Austin, TX, 2017 (First Place of TAU Timing Analysis Contest)
89. **[DAC'16]** Tsung-Wei Huang, Martin D. F. Wong, D. Sinha, K. Kalafala, and N. Venkateswaran, "A Distributed Timing Analysis Framework for Large Designs," *ACM/IEEE Design Automation Conference (DAC)*, Austin, TX, 2016
90. **[ICCAD'15]** Tsung-Wei Huang and Martin D. F. Wong, "OpenTimer: A High-performance Timing Analysis Tool," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, TX, 2015 (**2024 ICCAD 10-year Retrospective Most Influential Paper Award**)
91. **[SLIP'15]** Tsung-Wei Huang and Martin D. F. Wong, "On Fast Timing Closure: Speeding Up Incremental Path-Based Timing Analysis with MapReduce," *IEEE/ACM International Workshop on System-level Interconnect Prediction (SLIP)*, CA, 2015
92. **[ISPD'15]** Tsung-Wei Huang and Martin D. F. Wong, "Accelerated Path-Based Timing Analysis with MapReduce," *ACM International Symposium on Physical Design (ISPD)*, Monterey, CA, 2015
93. **[ICCAD'14]** Tsung-Wei Huang, P.-C. Wu, and Martin D. F. Wong, "Fast Path-Based Timing Analysis for CPPR," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Jose, CA, 2014 (First Place of TAU Timing Analysis Contest)
94. **[ICCAD'14]** Tsung-Wei Huang, P.-C. Wu, and Martin D. F. Wong, "UI-Timer: An Ultra-Fast Clock Network Pessimism Removal Algorithm," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Jose, CA, 2014
95. **[SLIP'14]** Tsung-Wei Huang, P.-C. Wu, and Martin D. F. Wong, "UI-Route: An Ultra-Fast Incremental Maze Routing Algorithm," *IEEE/ACM International Workshop on System-level Interconnect Prediction (SLIP)*, San Francisco, CA, 2014
96. **[ICCAD'12]** S.-H. Yeh, J.-W. Chang, Tsung-Wei Huang, and Tsung-Yi Ho, "Voltage-Aware Chip-Level Design for Reliability-Driven Pin-Constrained EWOD Chips," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Jose, CA, 2012
97. **[ISPD'12]** Tsung-Wei Huang, J.-W. Chang, and Tsung-Yi Ho, "Integrated Fluidic-Chip Co-Design Methodology for Digital Microfluidic Biochips," *ACM International Symposium on Physical Design (ISPD)*, Napa, CA, 2012
98. **[ASP-DAC'12]** J.-W. Chang, Tsung-Wei Huang, and Tsung-Yi Ho, "An ILP-based Obstacle-Avoiding Routing Algorithm for Pin-Constrained EWOD Chips," *IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*, Sydney, Australia, 2012
99. **[ICCAD'11]** Tsung-Wei Huang, Tsung-Yi Ho, and K. Chakrabarty, "Reliability-Oriented Broadcast Electrode-Addressing for Pin-Constrained Digital Microfluidic Biochips," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, San Jose, CA, 2011
100. **[SOCC'11]** Tsung-Wei Huang, Yan-You Lin, J.-W. Chang, and Tsung-Yi Ho, "Recent Research and Emerging Challenges in the Designs and Optimizations for Digital Microfluidic Biochips," *IEEE System on Chip Conference (SOCC)*, 2011.
101. **[MWSCAS'11]** Tsung-Wei Huang, Yan-You Lin, J.-W. Chang, and Tsung-Yi Ho, "Chip-Level Design and Optimization for Digital Microfluidic Biochips," *IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2011.
102. **[SLIP'11]** P.-H. Yuh, C. C.-Y. Lin, Tsung-Wei Huang, Tsung-Yi Ho, C.-L. Yang, and Y.-W. Chang, "A SAT-Based Routing Algorithm for Cross-Referencing Biochips," *IEEE/ACM International Workshop on System-level Interconnect Prediction (SLIP)*, San Diego, CA, June 2011.
103. **[DAC'11]** Tsung-Wei Huang, H.-Y. Su, and Tsung-Yi Ho, "Progressive Network-Flow Based Broadcast Addressing for Pin-Constrained Digital Microfluidic Biochips," *ACM/IEEE Design Automation Conference (DAC)*, pp. 741—746, San Diego, CA, June 2011.

104. **[ICCAD'10]** Tsung-Wei Huang, S.-Y. Yeh, and Tsung-Yi Ho, "A Network-Flow Based Pin-Count Aware Routing Algorithm for Broadcast Electrode-Addressing EWOD Chips," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, pp. 425-431, San Jose, CA, 2010.
105. **[ISPD'10]** Tsung-Wei Huang and Tsung-Yi Ho, "A Two-Stage Integer-Linear-Programming Based Droplet Routing Algorithm for Pin-Constrained Digital Microfluidic Biochips," *ACM International Symposium on Physical Design (ISPD)*, pp. 201—208, San Francisco, CA, 2010.
106. **[ICCAD'09]** Tsung-Wei Huang, C.-H. Lin, and Tsung-Yi Ho, "A Contamination-Aware Droplet Routing Algorithm for Digital Microfluidic Biochips," *IEEE/ACM International Conference on Computer-aided Design (ICCAD)*, pp. 151—156, San Jose, CA, 2009.
107. **[ICCD'09]** Tsung-Wei Huang and Tsung-Yi Ho, "A Fast Routability- and Performance-Driven Droplet Routing Algorithm for Digital Microfluidic Biochips," *IEEE International Conference on Computer Design (ICCD)*, pp. 445—450, Lake Tahoe, CA, 2009

JOURNAL PUBLICATION

1. **[TODAES'26]** Che Chang, Boyang Zhang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, Yibo Lin, and Tsung-Wei Huang, "C-PathGen: An Efficient CPU-Parallel Critical Path Generation Algorithm," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, accepted, 2026
2. **[TQC'26]** Shui Jiang, Hengrui Chen, Tsung-Yi Ho, and Tsung-Wei Huang, "FlatDD: Parallel Quantum Circuit Simulation using Decision Diagram and Flat Array," *ACM Transactions on Quantum Computing (TQC)*, accepted, 2026
3. **[TCAD'26]** Chih-Chun Chang, Yi-Hua Chung, Wan-Luan Lee, Boyang Zhang, and Tsung-Wei Huang, "SSTA-X: GPU-Accelerated First-Order Block-Based Statistical Static Timing Analysis," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, accepted, 2026
4. **[TCAD'26]** Shui Jiang, Wen Cheng, Yi-Hua Chung, Tsung-Yi Ho, and Tsung-Wei Huang, "QDP: Worst-case Fidelity-aware Qubit Mapping and Routing using Dynamic Programming," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, accepted, 2026
5. **[FGCS'25]** Serhan Gener, Sahil Hassan, H. Umut Suluhan, Liangliang Chang, Chaitali Chakrabarti, Tsung-Wei Huang, Umit Ogras, and Ali Akoglu, "A portable framework with generalized runtime features for task graph execution and concurrent multi-application deployment on heterogeneous systems," *Future Generation Computer Systems (FGCS)*, 2025
6. **[TODAES'25]** Wan-Luan Lee, Dian-Lun Lin, Shui Jiang, Cheng-Hsiang Chiu, Yibo Lin, Bei Yu, Tsung-Yi Ho, and Tsung-Wei Huang, "G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner using Task Graph Parallelism," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2025
7. **[TCAD'23]** G. Guo, Tsung-Wei Huang, Y. Lin, Z. Guo, S. Yellapragada, and M. D. F. Wong, "A GPU-Accelerated Framework for Path-Based Timing Analysis," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 42, no. 11, pp. 4219-4232, Nov. 2023
8. **[TCAD'23]** Zizheng Guo, Tsung-Wei Huang, and Yibo Lin, "Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 32, no. 12, pp. 4973-4984, Dec. 2023
9. **[TPDS'22]** Dian-Lun Lin and Tsung-Wei Huang, "Accelerating Large Sparse Neural Network Inference using GPU Task Graph Parallelism," *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, vol. 33, no. 11, pp. 3041—3052, Nov 2022
10. **[TPDS'22]** Tsung-Wei Huang, Dian-Lun Lin, Chun-Xun Lin, and Yibo Lin, "Taskflow: A Lightweight Parallel and Heterogeneous Task Graph Computing System," *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, vol. 33, no. 6, pp. 1303—1320, June 2022
11. **[TCAD'22]** Zizheng Guo, Mingwei Yang, Tsung-Wei Huang, and Yibo Lin, "A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 41, no. 10, pp. 3466—3478, Oct. 2022

12. **[JMIR'22]**Jia-Ruei Yu, Chun-Hsien Chen, Tsung-Wei Huang, Jang-Jih Lu, Chia-Ru Chung, Ting-Wei Lin, Min-Hsien Wu, Yi-Ju Tseng, Hsin-Yao Wang, "Energy Efficiency of Inference Algorithms for Medical Datasets: A Green AI study," *Journal of Medical Internet Research (JMIR)*, vol. 24, no. 1, Jan. 2022
13. **[TCAD'22]** Tsung-Wei Huang, Dian-Lun Lin, Yibo Lin, and Chun-Xun Lin, "Taskflow: A General-purpose Parallel and Heterogeneous Task Programming System," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 41, no. 5, pp. 1448—1452, May 2022
14. **[DAT'21]** Tsung-Wei Huang, Chun-Xun Lin, and Martin. D. F. Wong, "OpenTimer v2: A Parallel Incremental Timing Analysis Engine," *IEEE Design and Test (DAT)*, vol. 38, no. 2, pp. 62—68, April 2021
15. **[TCAD'21]** Tsung-Wei Huang, Yibo Lin, Chun-Xun Lin, Guannan Guo, and Martin. D. F. Wong, "Cpp-Taskflow: A General-purpose Parallel Task Programming System at Scale," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 40, no. 8, pp. 1687—1700, Aug. 2021
16. **[TCAD'21]** Tsung-Wei Huang, Guannan Guo, Chun-Xun Lin, and Martin. D. F. Wong, "OpenTimer v2: A New Parallel Incremental Timing Analysis Engine," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 40, no. 4, pp. 776—789, April, 2021
17. **[TCAD'18]** Tsung-Wei Huang, Chun-Xun Lin, and Martin D. F. Wong, "DtCraft: A High-performance Distributed Execution Engine at Scale," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 38, no. 6, pp. 1070—1083, June 2018
18. **[TCAD'16]** Tsung-Wei Huang and Martin D. F. Wong, "UI-Timer 1.0: An Ultra-Fast Path-Based Timing Analysis Algorithm for CPPR," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 35, no. 11, pp. 1862—1875, Nov. 2016
19. **[TCAD'14]** S.-H. Yeh, J.-W. Chang, Tsung-Wei Huang, S.-T. Yu, and Tsung-Yi Ho, "Voltage-Aware Chip-Level Design for Reliability-Driven Pin-Constrained EWOD Chips," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 33, no.9, pp. 1302—1315, Sep. 2014.
20. **[TCAD'13]** J.-W. Chen, C.-L. Hsu, L.-C. Tsai, Tsung-Wei Huang, and Tsung-Yi Ho, "An ILP-Based Routing Algorithm for Pin-Constrained EWOD Chips with Obstacle Avoidance," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 32, no.11, pp. 1655—1667, Nov. 2013.
21. **[TCAD'13]** Y.-H. Chen, C.-L. Hus, Tsung-Wei Huang, and Tsung-Yi Ho, "A Reliability-Oriented Placement Algorithm for Reconfigurable Digital Microfluidic Biochips using 3D Deferred Decision-Making Technique," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 32, no. 8, pp. 1151—1162, Aug. 2013.
22. **[TCAD'13]** J.-W. Chang, S.-H. Yeh, Tsung-Wei Huang, and Tsung-Yi Ho, "Integrated Fluidic-Chip Co-Design Methodology for Digital Microfluidic Biochips," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 32, no 2, pp. 216—227, Feb. 2013.
23. **[TCAD'11]** Tsung-Wei Huang, S.-Y. Yeh, and Tsung-Yi Ho, "A Network-Flow Based Pin-Count Aware Routing Algorithm for Broadcast-Addressing EWOD Chips," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 30, no. 12, pp. 1786—1799, Dec. 2011.
24. **[TCAD'11]** Tsung-Wei Huang and Tsung-Yi Ho, "A Two-Stage Integer-Linear-Programming Based Droplet Routing Algorithm for Pin-Constrained Digital Microfluidic Biochips," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 30, no. 2, pp. 215—228, Feb. 2011.
25. **[TCAD'10]** Tsung-Wei Huang, C.-H. Lin, and Tsung-Yi Ho, "A Contamination-Aware Droplet Routing Algorithm for the Synthesis of Digital Microfluidic Biochips," *IEEE Transactions on Computer-aided Design of Integrated Circuits and Systems (TCAD)*, vol. 29, no. 11, pp. 1682—1695, Nov. 2010.

PATENT

Incremental Common Path Pessimism Analysis USA-14/946043
Tsung-Wei Huang, K. Kalafala, D. Sinha, and N. Venkateswaran

Distributed Timing Analysis of a Partitioned Integrated Circuit Design USA-9916405B2
Tsung-Wei Huang, K. Kalafala, D. Sinha, and N. Venkateswaran

TALK

1. "Dynamic Asynchronous Tasking with Dependencies," CppOnline, April 2026
2. "Dynamic Asynchronous Tasking with Dependencies," CppCon, Sep 2025
3. "Taskflow: A General-purpose Task-parallel Programming System," Synopsys, Aug 2025
4. "An Efficient Implementation of Parallel BFS Algorithm," PPoPP FCPC, March 2025
5. "Intelligent High-performance Computing," Arizona State University, Oct 2024
6. "Scientific Research and Grant Writing," DAC Early Career Workshop, June 2024
7. "Taskflow: A General-purpose Task-parallel Programming System," Rapid Silicon, June 2024
8. "Intelligent High-performance Computing," University of Colorado Boulder, June 2024
9. "Intelligent High-performance Computing," University at Buffalo, May 2024
10. "Taskflow: A General-purpose Task-parallel Programming System," Cpp Rusia, May 2024
11. "Taskflow: A General-purpose Task-parallel Programming System," UCLA VAST Lab, April 2024
12. "Taskflow: A General-purpose Task-parallel Programming System," Cadence, April 2024
13. "Taskflow: A General-purpose Task-parallel Programming System," UC Santa Cruz, March 2024
14. "Taskflow: A General-purpose Task-parallel Programming System," FlexCompute Inc, Feb 2024
15. "Taskflow: A General-purpose Task-parallel Programming System," Chinese Univ of HK, Jan 2024
16. "Taskflow: A General-purpose Task-parallel Programming System," China Univ of Petroleum, Oct 2023
17. "Quantum Circuit Simulation with Incrementality," University at Buffalo, June 2023
18. "Intelligent High-performance Computing," Technical University of Munich, June 2023
19. "Taskflow: A General-purpose Task-parallel Programming System," Univ of Notre Dame, May 2023
20. "A General-purpose Task-parallel Programming System," Cruise LLC, April 2023
21. "Performance Portability and Optimization using Machine Learning," PNNL-Utah Seminar, Nov 2022
22. "Intelligent Heterogeneous Parallelism," ACCESS-CEDA Seminar Series, Hong Kong, Sep 2022
23. "Intelligent Heterogeneous Parallelism," CS Department, UC Merced, Sep 2022
24. "Programming System for Building High-performance CAD Applications," Google X, Sep 2022
25. "A General-purpose Parallel and Heterogeneous Task Programming System," AMD, Aug 2022
26. "A GPU Acceleration Flow for RTL Simulation with Batch Stimulus," Invited Talk, IWLS, July 2022
27. "Intelligent Heterogeneous Computing," AMD Research, June 2022
28. "Intelligent Heterogeneous Computing," ECE Department, Johns Hopkins University, March 2022
29. "Intelligent Heterogeneous Computing," ECE Department, Stevens Institute of Technology, 2022
30. "Intelligent Heterogeneous Computing," ECE Department, University of Minnesota, Feb 2022
31. "Taskflow: A General-purpose Heterogeneous Task Programming System," IXPUG, 2021
32. "cudaFlow: A Modern C++ Programming Model for GPU Task Graph Parallelism," CppCon, 2021
33. "A General-purpose Heterogeneous Task Computing System," Chinese Univ of HK, Aug 2021
34. "HeteroTime: Accelerating Static Timing Analysis with GPUs," Nvidia Research, June 2021
35. "Taskflow: A Lightweight Heterogeneous Task Programming System," CPPNow, 2021
36. "GPU-Accelerated Static Timing Analysis and Beyond," GTC, April 2021
37. "Machine Learning-enabled System for EDA," VLSI-DAT, April 2021
38. "GPU-Accelerated Static Timing Analysis," UC Santa Cruz, EDA Seminar, Feb 2021
39. "A General-purpose Heterogeneous Task Programming System," CIE/USA-GNYC, Oct 2020
40. "Taskflow: Parallel and Heterogeneous Task Programming in C++," C++ Meetup, Oct 2020
41. "Taskflow: A General-purpose Heterogeneous Task Programming System," CppIndia, Oct 2020
42. "Taskflow: A General-purpose Heterogeneous Task Programming System," MUC++, Oct 2020
43. "Programming Systems for Parallelizing VLSI CAD and Beyond," VLSI-DAT, April 2020
44. "A General-purpose Heterogeneous Task Programming System at Scale," ORNL, March 2020
45. "Growing Your Open-Source Projects," WOSET at IEEE/ACM ICCAD, November 2019
46. "Essential Building Blocks for Creating an Open-source EDA Project," IEEE/ACM DAC, June 2019
47. "Task-based Parallel Programming using Modern C++," CSL Social Hour, Sep 2018

48. "Distributed Timing Analysis in 100 Lines of Code," VSD webinar, May 2018
49. "DtCraft: A High-performance Distributed Execution Engine at Scale," CSLSC, UIUC, 2018
50. "OpenTimer: An open-source high-performance timing analysis tool," ORCONF, Italy, 2016
51. "Distributed Timing Analysis: Framework and Systems," Cadence, Austin, June 2016
52. "OpenTimer: A High-performance Timing Analysis Tool," Invited Talk, ICCAD, 2015
53. "Fast Path-based Timing Analysis," Invited Talk, ICCAD, 2014

PH.D. STUDENTS

Shui Jiang (co-supervised with Prof. Tsung-Yi Ho) — Fall 2022 – Summer 2026
Thesis: A System Stack for High-Performance Quantum Circuit Simulation

Wan-Luan Lee — Spring 2022 – Dec 2025
Thesis: GPU-accelerated Graph Partitioning Algorithms in VLSI Design — UW-Madison ECE PhD

Cheng-Hsiang Chiu — Fall 2020 – Summer 2025
Thesis: Asynchronous Many-task Systems with Intelligent Scheduling — UW-Madison ECE PhD

Jie Tong (co-supervised with Prof. Umit Y. Ogras) — Fall 2023 – Summer 2025
Thesis: Towards Scalable and Efficient Parallel RTL and Network Simulation Systems — UW-Madison ECE PhD

Dian-Lun Lin — Spring 2020 – Aug 2024
Thesis: Task-parallel Heterogeneous Programming System for Logic Simulation — UW-Madison ECE PhD

Guannan Guo (co-supervised with Prof. Martin Wong at UIUC) — Aug 2018 – Dec 2023
Thesis: Parallel and Heterogeneous Computing for Static Timing Analysis — UIUC ECE PhD

Che Chang — Spring 2022 – present
Thesis: TBD

Chih-Chun Chang — Spring 2023 – present
Thesis: TBD

Boyang Zhang — Spring 2023 – present
Thesis: TBD

Yi-Hua Chung — Fall 2023 – present
Thesis: TBD

Aditya Das Sarma — Fall 2024 – present
Thesis: TBD

Zhengxiong Li (co-supervised with Prof. Umit Y. Ogras) — Fall 2025 – present
Thesis: TBD

Zixuan Zhou — Fall 2026 – present
Thesis: TBD

INDUSTRIAL EXPERIENCE

Software Engineer, High-performance Computing Group, Citadel, IL — May 2017 – Aug 2017
Developed machine learning benchmarks and optimization tips for financial workloads

Software Engineer, Timing Analysis Group, IBM, NY — May 2015 – Aug 2015
Developed a distributed timing analysis prototype atop Einstimer

Software Engineer, Timing Analysis Group, Mentor Graphics, CA — May 2014 – Aug 2014
Developed optimization algorithms for tag-based incremental timing analysis

SERVICE

Grant Review Panelist

Evaluated grant proposals based on intellectual merit and broader impacts

- NSF Office of Advanced Cyberinfrastructure (OAC)
- NSF Principles and Practice of Scalable Systems (PPoSS)
- NSF Cyber-physical Systems (CPS)
- NSF Software and Hardware Foundations (SHF)
- DOE Advanced Scientific Computing Research (ASCR)
- DOE High Energy Physics (HEP)

Workershop Organization

Organized workshops and meetings to foster collaboration in CAD, HPC, and Quantum

- NSF FuSe Workshop on Intelligent Photonic Materials, Devices, and Systems, 2024
- NSF FuSe Workshop on Quantum Computing and Systems, 2024
- ACM/IEEE DAC Open-source Bird of Feather, 2024

Chair/Co-chair

Engaged widespread contributions to solving cutting-edge HPC and CAD problems

- IEEE/ACM ICCAD CAD Contests, 2020–2023
- ACM SIGDA CADathlon International Programming Contest, 2018–2021
- ACM TAU Timing Analysis Contest, 2018

Program Committee

Selected top-quality papers and organized conference programs

- ACM International Symposium on Computer Architecture (ISCA)
- ACM International Conference on Arch Support for Programming Languages and OS (ASPLOS)
- IEEE International Symposium on High-Performance Computer Architecture (HPCA)
- IEEE/ACM Design, Automation and Test in Europe Conference (DATE)
- IEEE/ACM International Conference on Computer-aided Design (ICCAD)
- IEEE/ACM International Symposium on Microarchitecture (MICRO)
- ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD)
- IEEE Computer Society Annual Symposium on VLSI (ISVLSI)
- IEEE International Parallel and Distributed Processing Symposium (IPDPS)
- ACM/IEEE International Conference for High Performance Computing (SC)
- ACM/IEEE Design Automation Conference (DAC)
- ACM International Workshop on Timing Issues (TAU)
- IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)
- IEEE International Conference on Computer Design (ICCD)
- International Workshop on Logic Synthesis (IWLS)
- C++ Conference (CppCon)

Editorship

Managed peer-review processes and recommended what gets published

- Guest editor, Special Issue of VLSI Integration, 2022

Journal Reviewers

Evaluated submitted papers and recommended acceptance/rejection

- IEEE Transactions on Parallel and Distributed Computing Systems (TPDS)
- IEEE Transactions on Computer-aided Design for Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Very Large-scale Integration (TVLSI)
- IEEE Transactions on Circuits and Systems (TCAS)
- IEEE Transactions on Big Data (TBD)
- IEEE Transactions on Architecture and Code Optimization (TACO)
- ACM Transaction son Design Automation of Electronic Systems (TODAES)
- ACM Journal of Emerging Technologies in Computing Systems (JETC)

- VLSI Integration Journal
- Concurrency and Computation: Practice and Experience
- Software X
- Journal of Computational Science

Departmental Committee at the University of Wisconsin at Madison
Helped the ECE department enhance various research and teaching programs

- Award Committee, 2026–Now
- Tenure and Promotion Committee, 2026–Now
- Faculty Search Committee, 2024–2025
- Undergraduate Advisory Committee, 2024–Now
- Accelerated MS Committee, 2024–Now
- Graduate Student and Admission Committee, 2023–2024

Departmental Committee at the University of Utah
Helped the ECE department enhance various research and teaching programs

- Graduate Student and Admission Committee, 2021–2023
- University of Utah Asia Campus Committee, 2021–2023
- University of Utah Asia Campus Students Summer Visit Program, 2021
- University of Utah Asia Campus faculty recruiting committee, 2021–2023
- Artificial Intelligence and Data-science faculty recruiting committee, 2020

MISCELLANEOUS

Citizenship: Taiwan
Hobby: Piano playing, hiking, camping